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TOILET SEAT ASSEMBLY WITH ELEVATED RING

Abstract

A toilet includes a bowl having an upper surface and a toilet seat coupled to the upper surface. The toilet seat includes a body including a top surface, an inner wall extending downward from an inner periphery of the body, and a lip extending downward from a bottom surface of the inner wall.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority benefit of Provisional Application No. 63/563,791 (Docket No. 010222-24010A) filed on Mar. 11, 2024, which is hereby incorporated by reference in its entirety.

FIELD
[0002] The present disclosure relates generally to toilets. More specifically, the present disclosure relates to a toilet seat assembly including an elevated ring.
BACKGROUND
[0003] Generally, elevated toilet seats may improve toilet accessibility and/or user comfort. However, there is an increased risk of waste soiling an underside of the seat or egressing between the seat and a bowl when an elevated seat is used. Particularly, the risk of waste soiling an underside of the seat or egressing may be most significant for male users sitting on the seat. Accordingly, there is need for an elevated toilet seat that reduces the risks of waste soiling an underside of the seat and/or egressing between the seat and bowl.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] Objects, features, and advantages of the present disclosure should become more apparent upon reading the following detailed description in conjunction with the drawing figures, in which:
[0005] FIG. **1** illustrates a perspective view of a toilet in accordance with one example of the present disclosure. Specifically, FIG. **1** illustrates a toilet including a tank in accordance with one example of the present disclosure.
[0006] FIG. **2** illustrates a perspective view of a toilet in accordance with one example of the present disclosure. Specifically, FIG. **2** illustrates a tankless toilet in accordance with one example of the present disclosure.
[0007] FIG. **3** illustrates a perspective view of a toilet seat in accordance with one example of the present disclosure.
[0008] FIG. **4** illustrates a cross sectional view of a toilet seat taken along section line AA of FIG. **3** in accordance with one example of the present disclosure.
[0009] FIG. **5** illustrates a cross sectional view of a toilet seat taken along section line BB of FIG. **3** in accordance with one example of the present disclosure.
[0010] FIG. **6** illustrates a cross sectional view of a toilet seat taken along section line CC of FIG. **3** in accordance with one example of the present disclosure.
[0011] FIG. **7** illustrates a cross sectional view of a toilet seat taken along section line DD of FIG. **3** in accordance with one example of the present disclosure.
[0012] FIG. **8** illustrates a perspective view of a toilet seat in accordance with one example of the present disclosure.
[0013] FIG. **9** illustrates a perspective view of toilet seat in accordance with one example of the present disclosure.
[0014] FIG. **10** illustrates a cross sectional view of a toilet seat taken along section line EE of FIG. **9**.
[0015] FIG. **11** illustrates a toilet in accordance with one example of the present disclosure.
[0016] FIG. **12** illustrates a partial perspective view of a toilet in accordance with one example of the present disclosure.
[0017] FIG. **13** illustrates an exploded view of a hinge assembly in accordance with one example of the present disclosure.
DETAILED DESCRIPTION
[0018] Described herein are toilet seats configured to prevent waste from soiling an underside of the seat and egressing between a seat and rim during use. Additionally, described herein are toilet seat assemblies and toilets including a seat configured to prevent waste from soiling an underside of the seat and egressing between a seat and rim during use. Specifically, described herein are elevated toilet seats including a body having a top surface configured to support a user, an inner

wall extending downward from an inner periphery of the body, and an outer wall extending downward from an outer periphery of the body. According to the present disclosure, the inner wall may include a stepped geometry configured to disrupt the adherence of liquid waste (e.g., urine) flowing along the seat. Specifically, electrostatic forces (e.g., attraction due to opposing charges) may cause liquid waste to adhere to and flow or travel along a surface or surfaces of the seat. The stepped geometry included in the inner wall may advantageously disrupt the adherence of a flow of liquid waste along a bottom surface of the inner wall, causing the liquid waste to fall downwards (e.g., into a toilet bowl) and preventing the liquid waste from following (e.g., continuing to adhere to) a contour of the seat and subsequently soiling a bottom surface of the seat and/or egressing between the seat and rim. According to some examples of the present disclosure the stepped geometry may include a lip extending downward from a bottom surface of the inner wall, such that a bottom surface of the wall forms an overhang disposed between an inner surface of the inner wall and the lip. According to some examples of the present disclosure, the inner wall may include a landing and a step extending below the landing.

[0019] FIGS. **1** and **2** illustrate exemplary embodiments of toilets that may include a toilet seat according to the present disclosure. The seats **131**, **231** described hereinafter and included in the toilets **100**, **200**, respectively, may be a toilet seat according to the present disclosure.

[0020] Referring to FIG. **1** a toilet **100** including a base **110** (e.g., a pedestal, bowl, etc.) and a tank **120** is shown. The base **110** is configured to be attached to another object such as a drainpipe, floor, or another suitable object. The base **110** includes a bowl **111**, a sump (e.g., a receptacle) disposed below the bowl **111**, and a trapway fluidly connecting the bowl **111** to a drainpipe or sewage line. The tank **120** may be supported by the base **110**, such as an upper surface of a rim **115**. The tank **120** may be integrally formed with the base **110** as a single unitary body. In other embodiments, the tank **120** may be formed separately from the base **110** and coupled (e.g., attached, secured, fastened, connected, etc.) to the base **110**. The toilet **100** may further include a tank lid **122** covering an opening and inner cavity in the tank **120**. The toilet **100** may include a seat assembly **130** including a seat **131** and a seat cover **132** rotatably coupled to the base **110**. The toilet **100** may further include a hinge assembly **135**.

[0021] Referring to FIG. **2**, a tankless toilet **200** is shown. The toilet **200** includes a base **210** and a seat assembly **230** coupled to the base. The base **210** includes a bowl **211**, a sump disposed below the bowl **211**, and a trapway fluidly connecting the bowl **211** to a drainpipe or sewage line. The toilet **200** includes a waterline **240** that supplies the toilet **200** with water. The toilet **200** may further include a seat assembly **230** including a seat **231** and a seat cover **232** rotatably coupled to the base **210**. The toilets **100** and **200** of FIGS. **1** and **2** are provided herein as non-limiting examples of toilets that may be configured to utilize aspects of the present disclosure.

[0022] Referring to FIG. **3**, a perspective view of a toilet seat **300** is illustrated in accordance with one example of the present disclosure. According to some examples of the present disclosure, the toilet seat **300** may include a body **310**, an inner wall **320** extending downward from an inner periphery of the body **310**, and an outer wall **330** extending downward from an outer periphery of the body **310**. The seat **300** may include a front **351**, a back **352**, a first (e.g., left) side **353**, and a second (e.g., right) side **354** opposite the first side **353**.

[0023] The body **310** may include a top surface **311** configured to support a user sitting on the seat **300** and a bottom surface **312** opposite the top surface. According to some examples, as shown in FIG. **3**, the body **310** (e.g., and the top and bottom surfaces **311**, **312**) may have an oblong, oval, egg, or elliptical band or ring shape. As illustrated in FIG. **3**, a hole or opening **340** may extend through the body **310** of the seat **300**. An inner periphery **341** (e.g., inner perimeter, inner circumference) of the body **310** may define a shape and/or size of the opening **340** extending through the body **310**.

[0024] In some examples, as illustrated in FIG. **3**, the body **310** (and the top and bottom surfaces **311**, **312**) may be narrower at a front **351** of the seat **300** than the body **310** (and top and bottom surfaces **311**, **312**) at the back **352** of the seat **300**. Similarly, the body **310** (and top and bottom surfaces **311**, **312**) may be wider at a back **352** of the seat **300** than the body **310** (and top and bottom surfaces **311**, **312**) at a front **351** of the seat **300**.

[0025] In some examples, as illustrated in FIG. **3**, the body **310** (and the top surface **311**) may be inclined. Specifically, in some examples, as illustrated in FIG. **3**, an inner periphery of the body **310** (and the top surface **311**) may be disposed below an outer periphery of the body **310** (and the top surface **311**). In other examples, the body **310** (and top surface **311**) may be horizontal (e.g., substantially horizontal, level).

[0026] Still referring to FIG. **3**, the toilet seat **300** may include an inner wall **320** extending or protruding downward from the inner periphery **341** of the body **310**. According to some examples, the inner wall **320** may extend downward from a bottom surface **312** of the body **310**. In some examples, as illustrated in FIGS. **3** and **4**, the inner wall **320** may extend downward at an angle (i.e., with respect to a vertical axis). For example, a bottom of the inner wall **320** may be disposed closer (e.g., radially, horizontally) to a center of the opening **340** than a top of the inner wall **320**. In other examples, the inner wall **320** may be vertical (e.g., substantially vertical), such that a top and bottom of the inner wall are disposed the same distance from the center of the opening **340**. According to some examples, as illustrated in FIG. **3**, the inner wall **320** may extend downward along an entirety of the inner periphery **341** of the body **310**. According to other examples, the inner wall **320** may extend downward along part of (e.g., less than all of) the inner periphery **341** of the body **310**.

[0027] According to the present disclosure, the inner wall **320** may further include stepped geometry configured to prevent liquid waste from soiling an under side (e.g., bottom side, bottom surface) of the toilet seat and egressing (e.g., leaving, exiting) the toilet by traveling between the seat **300** and a rim or upper surface of a pedestal or base (e.g., **110**, **210**). For example, as described hereinafter in greater detail with respect to FIGS. **4** and **5**, the inner wall **320** may include a lip **321** extending or protruding downward from a bottom surface **322** of the inner wall **320**. The stepped geometry may be configured to prevent liquid waste from adhering to and/or soiling one or more of a bottom surface **322** of the inner wall **320**, a lip **321** of the inner wall **320**, an outer surface of the inner wall **320**, a bottom surface **312** of the body **310**, an inner surface of the outer wall **330**, a bottom surface **331** of the outer wall **330**, and the like.

[0028] The toilet seat **300** may further include an outer wall **330** extending or protruding downward from an outer periphery **342** (e.g., outer perimeter, outer circumference) of the body **310**. According to some examples, the outer wall **330** may extend downward from a bottom surface **312** of the body **310**. In some examples, as illustrated in FIGS. **3** and **4**, the outer wall **330** may extend downward at an angle (i.e., with respect to a vertical axis). For example, a bottom of the outer wall **330** may be disposed further (e.g., radially, horizontally) from a center of the opening **340** than a top of the outer wall **330**. In other examples, the outer wall **330** may be vertical (e.g., substantially vertical), such that a top and bottom of the outer wall are disposed the same distance from a center of the opening **340**. According to some examples, as best seen in FIG. **8**, the outer wall **330** may extend downward along part of (e.g., less than all of) the outer periphery **342** of the body **310**. In accordance with other examples, the outer wall **330** may extend downward along an entirety of the outer periphery **342** of the body **310**.

[0029] Referring generally to FIGS. **4** and **5**, cross sectional views of the toilet seat **300** are illustrated. Specifically, FIG. **4** illustrates a cross section of the toilet seat **300** taken along section line AA of FIG. **3** in accordance with one example of the present disclosure. FIG. **5** illustrates a partial cross sectional view of the toilet seat **300** taken along section line BB of FIG. **3** in accordance with one example of the present disclosure.

[0030] Referring to FIGS. **4** and **5**, the inner wall **320** may include stepped geometry configured to disrupt a flow of liquid waste along one or more of the inner surface **323** of the inner wall **320**, the bottom surface **322** of the inner wall **320**, or a lip **321** of the inner wall **320**. Specifically, the inner wall **320** may include a step or lip **321** extending or protruding downward from the bottom surface **322** of the inner wall **320**. According to the present disclosure, the lip **321** may extend downward from less than an entire width of the bottom surface **322** of the inner wall **320**. Specifically, in some examples, the lip **321** may extend downward from the bottom surface **322** along an outer edge or periphery of the bottom surface **322**. The outer edge or periphery of the bottom surface **322** may be an edge or periphery of the bottom surface **322** furthest from the opening **340**. Accordingly, the bottom surface **322** may form a landing or overhang **324** disposed between the step or lip **321** and an inner surface **323** of the inner wall **320**. In accordance with some examples, the landing or overhang **324** may be a planar surface of the inner wall **320**. The step or lip **321** may extend below the landing or overhang **324**. The landing or overhang **324** may be disposed between the lip **321** and one or both of: the inner surface **323** of the inner wall **320**; or the opening **340**. In some examples, as shown in FIG. **5**, a width **327** of the lip **321** may be smaller than a width **328** of the overhang **324**. In accordance with other examples, the width **327** of the lip and the width of the overhang **324** may be the same. In accordance with yet other examples, the width **327** of the lip **321** may be larger than the width of **328** of the overhang **324**. According to some examples, as illustrated in FIGS. **4** and **5**, a bottom edge **325** of the step or lip **321** may extend below a bottom surface **331** of the outer wall **330**. According to some examples of the present disclosure, as illustrated in FIGS. **4** and **5**, the bottom surface **322** of the inner wall **320** may be planar. For example, the bottom surface **322** of the inner wall **320** may be horizontal (e.g., substantially horizontal, level).

[0031] Still referring to FIGS. **4** and **5**, in some examples, the lip **321** may extend downward from the bottom surface **322** along less than an entire length or circumference of the bottom surface **322**. According to some examples, the lip **321** may only extend along a portion of (e.g., less than an entire circumference of) the inner wall **320** disposed at or near the front **351** of the seat **300**. For example, the lip **321** may extend from the bottom surface along less than half the length (e.g., circumference) of the bottom surface In other examples, the lip **321** may extend downward from the bottom surface **322** along an entire length (e.g., circumference) of the inner wall **320**.

[0032] According to some examples, as illustrated in FIGS. **4** and **5**, a height or distance (e.g., in the vertical direction) between the bottom surface **322** and/or overhang **324** and a distal end or bottom edge **325** of the lip **321** may vary along a length (e.g., circumference) of the inner wall **320**. Specifically, a vertical distance between the bottom edge **325** of the lip **321** and the bottom surface **322** and/or overhang **324** of the inner wall **320** and/or the body **310** may vary along the length (e.g., circumference) of the inner wall **320**. According to some examples of the present disclosure, as illustrated in FIGS. **4** and **5**, the lip **321** may have a largest height or extend downward a largest distance (e.g., from the bottom surface **322**, from the body **310**) at a front **351** of the inner wall **320**. Additionally, as illustrated in FIGS. **4** and **5**, according to some examples of the present disclosure, the height of the lip **321** or the distance the lip extends downward may gradually decrease along the length of the inner wall **320** from a largest height or largest distance at a front **351** of the inner wall **320** toward a back **352** of the inner wall **320** and/or a back **352** of the seat **300**.

[0033] According to some examples of the present disclosure, as illustrated in FIGS. **4** and **5**, the lip **321** may have a largest height or extend downward a largest distance (e.g., from the bottom surface **322**, from the body **310**) at a center of the inner wall **320**. Additionally, as illustrated in FIGS. **4** and **5**, according to some examples of the present disclosure, the height of the lip **321** or the distance the lip extends downward may gradually decrease along the length of the inner wall **320** from a largest height or largest distance at a center of the inner wall **320** toward the first side **353** and the second side **354** of the inner wall **320** and/or the first side **353** and second side **354** of the seat **300**.

[0034] According to some examples, as illustrated in FIGS. **4** and **5**, the height of the lip **321** and/or a distance the lip **321** extends downward may gradually decrease (e.g., toward the back **352** and/or sides **353**, **354**) until the lip **321** is coextensive with a bottom surface **322** of the inner wall **320**.

[0035] Referring to FIG. **6**, a cross section view of the front **351** of the toilet seat **300** taken along section line CC of FIG. **3** is illustrated in accordance with one example of the present disclosure. Referring to FIG. **7**, a cross section view of the back **352** of the toilet seat **300** taken along section line DD of FIG. **3** is illustrated in accordance with one example of the present disclosure.
[0036] Referring to FIGS. **6** and **7**, the toilet seat **300** may further include one or more (e.g., at least one) leg **360**. Each leg **360** may extend or protrude downward from a bottom surface **312** of the body **310**. In accordance with some examples of the present disclosure, as illustrated in FIGS. **6** and **7**, each of the legs **360** may extended downward beyond a bottom surface **322** of the inner wall **320**, beyond a bottom edge **325** of the lip **321**, and below a bottom surface **331** of the outer wall **330**. Each leg **360** may be configured to contact an upper surface of a base or pedestal or rim of (e.g., **315**) of a toilet and support the body **310**, inner wall **320**, and outer wall **330**. Specifically, in some examples, as described hereinafter in more detail with respect to FIG. **11**, the legs **360** may be configured to support the body **310**, inner wall **320**, and outer wall **330** such that they are elevated above an upper surface of the base or a rim (e.g., **315**) of a toilet.
[0037] In some examples, as illustrated in FIGS. **6** and **7**, each of the legs **360** may be disposed between the inner wall **320** and the outer wall **330**. Specifically, each leg **360** may extend from a bottom surface **312** of the body **310** between the inner wall **320** and the outer wall **330**. According to some examples, as illustrated in FIGS. **6** and **7**, the toilet seat **300** may include four legs **360**. Specifically, as illustrated in FIG. **6**, the seat **300** may include two legs **360** extending from the bottom surface **312** at a front **351** of the seat **300**. Additionally, as illustrated in FIG. **7**, the seat **300** may include two legs **360** extending from the bottom surface **312** at a back **352** of the seat **300**. However, the present disclosure is not limited thereto and any number of legs **360** may be included.
[0038] According to some examples, as illustrated in FIGS. **5** and **6**, each leg **360** may abut and/or be disposed directly adjacent to the outer wall **330**. According to some examples of the present disclosure, any one, any combination, or all of: the body **310**: inner wall **320**; lip **321**; outer wall **330**; or legs **360** may be integrally formed as a single unitary body. According to other examples of the present disclosure, one or more of the body **310**, inner wall **320**, lip **321**, outer wall **330**, and legs **360** may be separately formed and the body **310**, inner wall **320**, lip **321**, outer wall **330**, and/or legs **360** may be coupled or attached to one another to form the toilet seat **300**.
[0039] Referring to FIG. **8**, a perspective view of a toilet seat **300** is illustrated in accordance with one example of the present disclosure. As shown in FIG. **8**, the toilet seat **300** may include a pair of hinge tabs **345** extending or protruding from the body **310** (e.g., bottom surface **312**) and/or the outer wall **330**. The pair of hinge tabs **345** may be disposed at a back **352** of the seat **300**. As shown in FIG. **8**, one of the pair of hinge tabs **345** may be disposed at the back **352** of the seat **300** on each of the first side **353** and the second side **354** of the seat **300**.
[0040] As shown in FIG. **8**, the seat **300** may further include a pair of hinge openings **346**. Specifically, a hinge opening **346** may extend through each of the hinge tabs **345**. In some examples, as illustrated in FIG. **8**, a line extending through the central axis of each of the pair of hinge openings **346** may be coincident. Each of the pair of hinge openings may be configured to receive a portion of a hinge (e.g., hinge pin) for rotatably coupling the seat **300** to a base or pedestal of a toilet.
[0041] Still referring to FIG. **8**, the seat **300** may further include a hinge slot **347** configured to receive a portion of a hinge (e.g., hinge pin) for rotatably coupling the seat **300** to a base or pedestal of a toilet. According to some examples, as illustrated in FIG. **8**, the hinge slot **347** may be disposed at a back **351** of the seat **300**. As shown in FIG. **8**, the hinge slot **347** may be disposed between the pair of hinge tabs **345**. According to some examples of the present disclosure, each of the pair of hinge tabs **345** and the bottom surface **312** of the body **310** may define bounds of the hinge slot **347**. Coupling of a hinge assembly to the toilet seat **300** is described hereinafter in greater detail with respect to FIG. **12**.
[0042] Referring to FIG. **9**, a perspective view of a toilet seat **400** is illustrated in accordance with

one example of the present disclosure. The toilet seat **400** may be the same or substantially similar to the toilet seat **300** described above. The toilet seat **400** may include a body **410**, an inner wall **420**, and an outer wall **430**. Each of the body **410**, inner wall **420**, and outer wall **430** may be the same or substantially similar to the body **310**, inner wall **320**, and outer wall **330**, respectively, as described above, except, the body **410**, inner wall **420**, and/or outer wall **430** may have different profiles or contours (e.g., shapes) than those described above. In accordance with some examples of the present disclosure, the seat **400** may further included one or more legs **360** extending or protruding downward from the body **410** of the seat **400**. In some examples, the seat **400** may include four legs **360**. Each of the legs **360** may be disposed between the inner wall **420** and the outer wall **430**.

[0043] Referring to FIG. **10**, a cross sectional view of the toilet seat **400** taken along section line EE of FIG. **9** is illustrated in accordance with one example of the present disclosure. As shown in FIG. **10**, the inner wall **420** may include a lip **421** having a bottom edge **425** extending downward from a bottom surface **422** of the inner wall **420**. The lip **421**, bottom edge **425**, and the bottom surface **422** may be the same as the lip **321**, bottom edge **325**, and bottom surface **422**, respectively, described above with respect to FIGS. **3-5**. As illustrated in FIG. **10**, the outer wall **430** may include a bottom surface **431**.

[0044] According to some examples of the present disclosure, as illustrated in FIG. **10**, each of the legs **360** may extend beyond or below a bottom surface **422** of the inner wall **420**, a bottom edge **425** of the lip **421**, and a bottom surface **431** of the outer wall **430**. Additionally, in some examples, as illustrated in FIG. **10**, the outer wall **430** may extend beyond or below a bottom edge **425** of the lip **421**. According to some examples, as illustrated in FIG. **10**, a bottom edge **425** of the lip may be horizontal (e.g., substantially horizontal, level).

[0045] Referring to FIG. **11**, a side view of a toilet **500** is illustrated in accordance with one example of the present disclosure. In some examples, as illustrated in FIG. **11**, the toilet **500** may include a toilet seat **501**, a base or pedestal **510**, a hinge assembly **520**, and a toilet seat cover **530**. The toilet seat **501** may be either of the toilet seats **300** and **400** described herein. For ease of explanation, the toilet **500** is described hereinafter as including the toilet seat **300**; however, the present disclosure is not limited thereto and the toilet seat **501** may also be the toilet seat **400** described above.

[0046] Referring to FIG. **11**, the pedestal or base **510** includes a bowl **511**, a receptacle or sump **512** disposed below the bowl **511**, and a trapway **515** configured to fluidly connect the bowl **511** to a drainpipe or sewage line. According to the present disclosure, the base **510** may further include a rim **513** extending around the bowl **511**. The base **510** may further include an upper surface **514** configured to support a tank (e.g., **120**) and the toilet seat **501**. Specifically, an upper surface **514** of the rim **513** may support the toilet seat **501**. In accordance with the present disclosure, the stepped geometry of the toilet seat **300**, specifically, the bottom surface **322** of the inner wall **320**, the overhang **324**, and/or the lip **321** may advantageously disrupt or stop the adherence of liquid waste flowing along the inner wall **320**, bottom surface **322**, and/or lip **321**, causing liquid waste to, for example, fall into the bowl **511**, preventing liquid waste from flowing between the inner wall **320** and the rim **513** (e.g., to an outer surface of the inner wall **320**, to a bottom surface **312** of the body **310**, to the outer wall **330**, out of the toilet **500**).

[0047] As shown in FIG. **11**, each of the legs **360** may be configured to contact or abut the upper surface **514** of the rim **513**. The legs **360** may be configured to contact the upper surface **514** of the rim **513** and support the body **310**, inner wall **320**, and outer wall **330** of the seat **300**. Specifically, the legs **360** may be configured to support the seat **300** such that the body **310**, inner wall **320**, and outer wall **330** are elevated or disposed above the upper surface **514** of the rim **513**.

[0048] According to some examples of the present disclosure, the legs **360** may be configured to support the body **310** such that the top surface **311** of the seat is disposed a predetermined distance above the upper surface **514** of the rim **513**. Specifically, the legs **360** may be configured to support the top surface **311** of the seat a predetermined distance above the upper surface **514** of the base **510** to comply with government regulations and sitting height requirements for those with disabilities. According to some examples, the legs **360** may support the body **310**, such that, a top surface **311** of the seat **300** is disposed more than 1 inch, more than 1.5 inches, more than 2 inches, more than 2.5 inches, or 3 or more inches above an upper surface **514** of the rim.

[0049] According to some examples, in conjunction with a height of the top surface **311** above the upper surface **514** of the rim **513** a predetermined leg **360** height may be required. According to some examples, each leg **360** may extend downward more than 1 inch, more than 1.5 inches, more than 2 inches, more than 2.5 inches, or 3 or more inches from the bottom surface **312** of the body **310**. Similarly, according to some examples, in conjunction with a height of the top surface **311** above the upper surface **514** of the rim **513** and a length of the legs **360**, the inner wall **320** and the outer wall **330** may extend downward a predetermined distance from the bottom surface **312** of the body **310**.

[0050] According to some examples, as illustrated in FIG. **11**, the inner wall **320** and the lip **321** may be disposed above the bowl **511** of the toilet **500**. Specifically, the inner wall **320** and the lip **321** may be disposed within the rim **513** of the base **510**, such that, when adherence of a flow of liquid waste along the inner surface **323** and/or bottom surface **322** of the wall is disturbed, the liquid waste falls into the bowl **511**.

[0051] As illustrated in FIG. **11**, the hinge assembly **520** may be coupled to the base **510**, the seat **501**, and the seat cover **530** and may rotatably couple to seat **501** and the seat cover **530** to the base **510**. The hinge assembly **520** is described below in greater detail with respect to FIGS. **12** and **13**.

[0052] Referring to FIG. **12**, a partial perspective view of a toilet **500** is illustrated in accordance with one example of the present disclosure. In accordance with some examples of the present disclosure, as illustrated in FIG. **12**, the hinge assembly **520** may include a pair of hinge towers **540** and a pin assembly **550**. The pair of hinge towers **540** may be coupled to the upper surface **514** of the base **510** and to the pin assembly **550**. According to some examples, as is described hereinafter with respect to FIG. **13**, the hinge towers **540** may be disposed below the pin assembly **550** and coupled to a bottom of the pin assembly **550**. According to some examples, the pair of hinge towers **540** may be coupled to and support the pin assembly **550**, such that, the pin assembly **550** is elevated above an upper surface **514** of the base **510**. The pin assembly **550** may be disposed above and extend between the pair of hinge towers **540**. Specifically, the pair of hinge towers **540** may support the pin assembly **550** above the upper surface **514** of the base **510**, such that an opening or gap **516** is formed or exists between the upper surface **514** of the base **510**, the pin assembly **550**, and the pair of hinge towers **540**. In accordance with some examples, a height of the hinge towers **540** may be determined in consideration of a seat height (e.g., vertical distance between the top surface **311** of the seat **300** and the upper surface **514** of the base **510**). The pin assembly **550** may rotatably couple the seat **501** and the seat cover **530** to the hinge towers **540** (and thus the upper surface **514** of the base **510** when the hinge towers **540** are coupled to the base **510**). According to some examples, as described hereinafter with respect to FIG. **13**, the pin assembly **550** may be disposed in the hinge slot **347** at the back **352** of the seat **300**.

[0053] Referring to FIG. **13**, an exploded view of a hinge assembly **520** is illustrated in accordance with one example of the present disclosure. As noted above and illustrated in FIG. **13**, the hinge assembly **520** may include a pair of hinge towers **540** and a pin assembly **550**. As shown in FIG. **13**, each of the hinge towers **540** may include a foot **541** configured to be coupled to the upper surface **514** of the base **510** and a shaft **542** (e.g., vertical shaft). According to some examples, the shaft **542** may extend in a vertical (e.g., substantially vertical) direction. In some examples, the shaft **542** may be inserted into a hole **543** (e.g., disposed in an upper surface of the foot **541**) and extend or protrude upward from the foot **541**. In other examples, the shaft **542** may be integrally formed with the foot **541** and extend or protrude upward from the foot **541**.

[0054] Still referring to FIG. **13**, in accordance with some examples, the pin assembly **550** may include a hollow channel **551**, a locking mechanism **552**, and a pair of dampers **553**. According to some examples, the hollow channel **551** may include a pair of tower coupling holes **554**. The pair of tower coupling holes **554** may be disposed along a bottom of the hollow channel **551**. Each of the pair of tower coupling holes **554** may be configured to receive a shaft **542** of one of the hinge towers **540** for coupling the hinge tower **540** and the pin assembly **550**. Further, the hollow channel **551** may include a cavity configured to receive the locking mechanism **552** and the pair of dampers **553**. According to some examples, the pin assembly **550** and specifically the hollow channel **551** may be disposed in the hinge slot **347** disposed at a back **352** of the toilet seat **300**.

[0055] In accordance with some examples of the present disclosure, each of the pair of dampers **553** may be disposed within (e.g., the cavity of) the hollow channel **551**. One of the pair of dampers **553** may be disposed at each of opposite ends of the hollow channel **551**. According to some examples, a portion of each of the pair of dampers **553** may extend or protrude out of an opening at a respective end of the hollow channel **551**. According to some examples, a portion of each of the dampers **553** extending outside of the hollow channel **551** may be inserted into a hinge opening **346** disposed on a respective side of the toilet seat **300**. Additionally, the portion of each damper **553** extending from the hollow channel **551** may also extend into openings formed in the seat cover **530**, rotatably coupling the seat cover **530** to the pin assembly **550**.

[0056] Referring to FIG. **13**, the locking mechanism **552** may be disposed within the hollow channel **551** and may be configured to selectively lock or maintain a position of the pair of hinge towers **540** and/or the pair of dampers **553** (i.e., and thus the toilet seat **300** and seat cover **530**) with respect to the hollow channel **551**. According to some examples, the locking mechanism **552** may include a button **555**. The button **555** may be disposed in a button opening **556** included in the hollow channel **551**. The locking mechanism **552** may further include one or more biasing members **557** (e.g., springs). In a first state, the biasing members **557** may apply (e.g., an axial) force on each of the pair of dampers **553** toward a respective end of the hollow channel **551**, maintaining a position of the dampers **553** with respect to the hollow channel **551** and maintaining a position of a portion of each of the dampers **553** within a respective hinge opening **346** included in the seat **300**. When the button **555** is pressed (e.g., depressed), the biasing members **557** may be compressed and each of the dampers **553** may be withdrawn from a respective hinge opening **346**, allowing the seat **300** and seat cover **530** to be removed from the hinge assembly **520**.

[0057] When a component, device, element, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element should be considered herein as being “configured to” meet that purpose or to perform that operation or function.

[0058] As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0059] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0060] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0061] The term “or,” as used herein, is used in its inclusive sense (and not in its exclusive sense) so that when used to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is understood to convey that an element may be either X, Y, Z; X and Y; X and Z; Y and Z; or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

[0062] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0063] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0064] It is important to note that the construction and arrangement of the system as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. A toilet seat comprising: a body including a top surface; an opening extending through the body, the opening defining an inner periphery of the body; an inner wall extending downward from the inner periphery of the body; a lip extending downward from a bottom surface of the inner wall; and an overhang formed by the bottom surface of the inner wall, the overhang disposed between the lip and the opening.

2. The toilet seat of claim 1, wherein the overhang and the lip are configured to disrupt adherence of liquid waste flowing along the toilet seat, causing liquid waste adhering to the toilet seat to fall.

3. The toilet seat of claim 1, wherein a distance the lip extends downward varies along a circumference of the bottom surface of the inner wall.

4. The toilet seat of claim 3, wherein the lip extends downward a largest distance at a front of the inner wall.

5. The toilet seat of claim 4, wherein the distance the lip extends downward gradually decreases

from the largest distance at the front of the inner wall toward a back of the inner wall.
6. The toilet seat of claim 5, wherein the distance the lip extends downward gradually decreases until the lip is coextensive with the bottom surface of the inner wall.
7. The toilet seat of claim 1, wherein a distance the lip extends downward gradually decreases from a largest distance at a middle of the inner wall toward a first side and a second side of the inner wall.
8. The toilet seat of claim 1, further comprising: an outer wall extending downward from an outer periphery of the body.
9. The toilet seat of claim 8, wherein the lip extends downward beyond a bottom surface of the outer wall.
10. The toilet seat of claim 8, wherein the outer wall extends downward beyond the lip.
11. The toilet seat of claim 8, further comprising: a leg extending downward from the body between the inner wall and the outer wall.
12. The toilet seat of claim 11, wherein the leg extends beyond the bottom surface of the inner wall, a bottom edge of the lip, and a bottom surface of the outer wall.
13. A toilet comprising: a bowl having an upper surface; and a toilet seat coupled to the upper surface, the toilet seat having a body including a top surface and an opening extending through the body, an inner wall extending downward from an inner periphery of the body, an overhang formed by a bottom surface of the inner wall, and a lip extending downward from the bottom surface of the inner wall.
14. The toilet of claim 13, wherein the overhang is disposed between the lip and the opening.
15. The toilet of claim 13, wherein the lip and the overhang are configured to disrupt adherence of liquid waste flowing along the toilet seat, causing the liquid waste to fall into the bowl.
16. The toilet of claim 13, further comprising: an outer wall extending downward from an outer periphery of the body; and at least one leg extending downward from the body between the inner wall and the outer wall.
17. The toilet of claim 16, wherein the at least one leg is configured to abut the upper surface of the bowl and support the toilet seat, such that the body, the outer wall, the inner wall, and the lip are elevated above the upper surface of the bowl.
18. The toilet of claim 16, wherein the at least one leg includes four legs.
19. A toilet seat comprising: a body; an opening extending through the body, the opening defining an inner periphery of the body; and an inner wall extending from the inner periphery of the body, the inner wall including a landing and a step extending below the landing, the landing and the step configured to disrupt adherence of liquid waste flowing along the inner wall, causing liquid waste flowing along the inner wall to fall from the inner wall.
20. The toilet seat of claim 19, further comprising: an outer wall extending downward from an outer periphery of the body; and at least one leg extending downward from the body between the inner wall and the outer wall.